מלגה 2

$(\neg P \vee Q) \vee ((P \to Q) \wedge Q) = P \to Q$ ⓐ

P	Q	$\neg P$	$P \to Q$	$\neg P \vee Q$	$(P \to Q) \wedge Q$	$(\neg P \vee Q) \vee ((P \to Q) \wedge Q)$
F	F	T	T	T	F	T
F	T	T	T	T	T	T
T	F	F	F	F	F	F
T	T	F	T	T	T	T

$(P \to (Q \vee R)) = (P \to Q) \vee (P \to R)$ ⓑ

P	Q	R	$P \to Q$	$P \to R$	$Q \vee R$	$P \to (Q \vee R)$	$(P \to Q) \vee (P \to R)$
F	F	F	T	T	F	T	T
F	F	T	T	T	T	T	T
F	T	F	T	T	T	T	T
F	T	T	T	T	T	T	T
T	F	F	F	F	F	F	F
T	F	T	F	T	T	T	T
T	T	F	T	F	T	T	T
T	T	T	T	T	T	T	T

$(A \wedge \neg B) \vee (A \wedge B) = A$ ⓖ

A	B	$\neg B$	$A \wedge \neg B$	$A \wedge B$	$(A \wedge \neg B) \vee (A \wedge B)$
F	F	T	F	F	F
F	T	F	F	F	F
T	F	T	T	F	T
T	T	F	F	T	T

$\neg Q \to \neg P = P \to Q$ ⓓ

P	Q	$\neg P$	$\neg Q$	$\neg Q \to \neg P$	$P \to Q$
F	F	T	T	T	T
F	T	T	F	T	T
T	F	F	T	F	F
T	T	F	F	T	T

$(P \wedge Q) \vee (\neg P \wedge R) \vee (Q \wedge R) = (P \wedge Q) \vee (\neg P \wedge R)$ ①

P	Q	R	$\neg P$	$Q \wedge R$	$\neg P \wedge R$	$P \wedge Q$	$(P \wedge Q) \vee (\neg P \wedge R) \vee (Q \wedge R)$	$(P \wedge Q) \vee (\neg P \wedge R)$
F	F	F	T	F	F	F	F	F
F	F	T	T	F	T	F	T	T
F	T	F	T	F	F	F	F	F
F	T	T	T	T	T	F	T	T
T	F	F	F	F	F	F	F	F
T	F	T	F	F	F	F	F	F
T	T	F	F	F	F	T	T	T
T	T	T	F	T	F	T	T	T

$A \vee (\neg A \wedge \neg B) = B \to A$ ⓘ

A	B	$\neg A$	$\neg B$	$\neg A \wedge \neg B$	$A \vee (\neg A \wedge \neg B)$	$B \to A$
F	F	T	T	T	T	T
F	T	T	F	F	F	F
T	F	F	T	F	T	T
T	T	F	F	F	T	T